

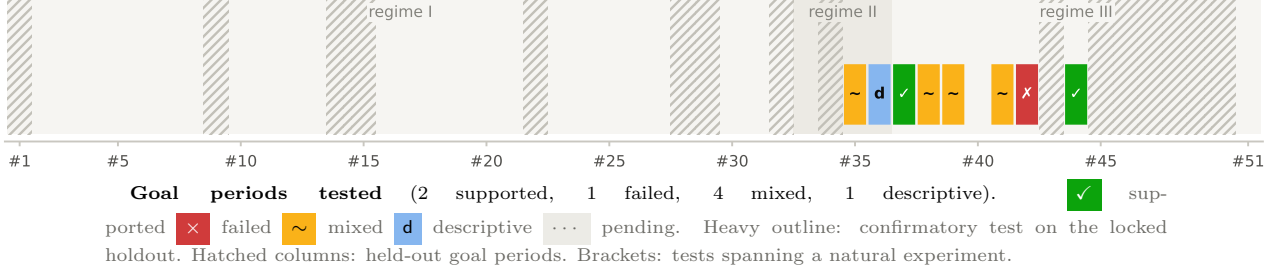
H107: Does the room split grow from zero (an instability) or appear at once (a hidden field)?

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 1/5

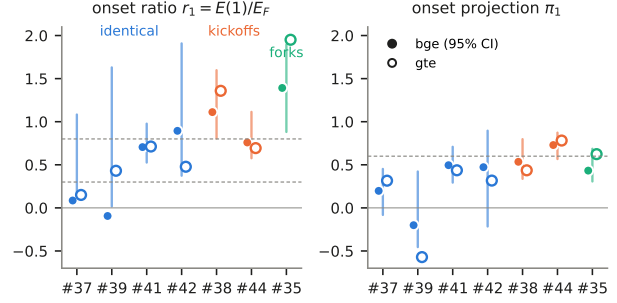
Bottom line. The room content split has no single onset shape: it grows from zero in two identical-kickoff weeks and is nearly full on day 1 in two others, its day-1 direction is half kept, and inherited repos set it only on one week's first day.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH337 asks whether H100's spontaneous #best/#rest content split is a true symmetry breaking or a hidden field. True symmetry breaking starts at zero on day 1 and grows, with a direction chosen by early fluctuations. A hidden field, such as the work each room inherits, gives a full split on day 1 along a direction the members' previous repos predict. H107 measures the day-1 split against the final one in 8 non-holdout two-room periods, with the inherited repo field as an explicit covariate. **Neither picture covers all four identical-kickoff weeks.** In #37 and #39 the split starts near zero and grows. In #41 and #42 it is at 70–90% of final size on day 1, but the day-1 direction is only half kept. Inherited repos set the day-1 direction in #42 only.



Observables. Onset ratio r_1 (left) and onset projection π_1 (right) per period; filled = bge with agent-bootstrap 95% CI, open = gte. Dashed lines: the SSB (0.3) and kill (0.8) thresholds; the positive-control threshold 0.6. Identical kickoffs (blue) split two ways: #37 and #39 start near zero, #41 and #42 near full size. Room kickoffs (orange) and forks (green) give full-size day-1 splits, but only #44 keeps its direction.

PHYSICAL MODEL

Model 11 (soft-spin $O(n)$, $n = 32$ whitened dimensions), with the inherited work as a model-10 Potts field. $\mathbf{x}_{i,b}$ is agent i 's style-residualized content in time bin b , centred on the bin and minus its leave-period-out constant. Δ is the room order parameter. $E(b)$ is its squared size in excess of random joint relabels. The inherited field $\mathbf{h}_r$ weights each repo's content vector $\mathbf{v}_\sigma$ by the members' pre-period commit shares $w_{i\sigma}$.

$$\Delta(b) = \bar{\mathbf{x}}_{\text{best}}(b) - \bar{\mathbf{x}}_{\text{rest}}(b), \quad C(b, b') = \Delta(b) \cdot \Delta(b') - \langle \cdot \rangle_\pi$$

$$r_1 = E(1)/E_F, \quad c_1 = C(1, F)/\sqrt{E(1)E_F}, \quad \pi_1 = C(1, F)/E_F$$

$$\mathbf{h}_r = \langle \sum_\sigma w_{i\sigma} \mathbf{v}_\sigma \rangle_{i \in r}, \quad f_{\text{repo}} = (\Delta^{(1)} \cdot \hat{\mathbf{u}})(\Delta^{(2)} \cdot \hat{\mathbf{u}})/S$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, this mostly says what not to infer. The room difference you see on day 1 of a week with identical instructions is a poor guide to the week: in some weeks it is absent and grows, and where it exists its direction is only about half kept ($c_1 \approx 0.5$). The work the members bring in rarely sets it, which weakens the “hidden field” explanation of H100's residual without proving true symmetry breaking. Day resolution cannot separate a field present at $t = 0$ from an instability that grows within two hours. This is a measurement of onset shape, not a lever.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P0 synthetic: size, step, SSB, repo power	E_F size 0.02; kill 75%, $\pi_1 \geq 0.6$ 100% (step); SSB 76% (slow); repo size 0.03–0.05, power 1.0	passed
P1 SSB clause $\geq 3/4$; kill $\geq 2/4$; median $\pi_1 \geq 0.5$	2/4 (#37, #39); 0/4; 0.33 (gte 0.32)	not met
P2 first half-day ≤ 0.3 in $\geq 3/4$	2/4 (#41 0.82, #42 2.38)	not met
P3 repo clause $\geq 2/4$; $\kappa_w > 0$	period 0/4, day 1 1/4 (#42: 0.34, p 0.002); $\kappa_w > 0$ 4/4	not met; met
P4 positive control $\pi_1 \geq 0.6$	#44 0.73; #38 0.53 (r_1 1.11)	failed (half)
P5 forks $\pi_1 \geq 0.6$	0.43 (gte 0.62)	failed
P6 #39 inherits #38 work	repo 0.00; carried content (gte 0.22)	mixed
P7 #41 recalls #39	z -4.5 (anti-aligned); #40 repos n.s.	failed (reversed)
HH337	2 supported, 1 failed, 1 mixed; P4 failed	mixed

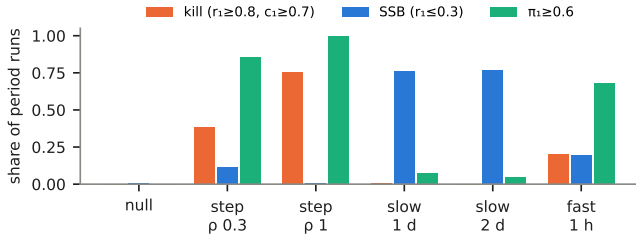
FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

Four identical-kickoff periods with 3–5 #best agents; r_1 CIs span 1.0–1.9 in #37, #39 and #42. A growth time below two active hours cannot be told from a field at $t = 0$. The positive control is half met: #38 has a full-size day-1 split whose direction is half kept, a day-1 transient (post hoc check at days 4–5: π_1 0.56). Repo vectors need same-regime statements naming the repo before the period, so #37's field rests on four days. The #41 anti-alignment and the reshuffle reading of #39 are post hoc.



Second figure. Synthetic validation on the real statement skeletons (40 worlds, 7 periods): share of period runs in which the HH kill, the SSB clause and $\pi_1 \geq 0.6$ fire. Step worlds trigger the kill and π_1 ; slow growth triggers the SSB clause; an instability with a growth time of about one hour looks like a step at day resolution.

ROUND 2

Hour-resolution onset on day 1 with hierarchical pooling across periods; the reshuffle onset on NE19 (#45, confirm C4); what the day-1 transient follows (operator messages vs the first repos touched).

- **Essence.** whether a room's content order is born from a field the room inherits or from an instability, and how fast it orients.
- **H107-R1** Hour resolution on day 1.
- **H107-R2** What sets the onset shape.
- **H107-R3** Contemporaneous work field (H100-R3): the room's period-P repo shares as a Potts field; does the final split lie along what each room builds?
- **H107-R4** Day-1 transient mechanism.

GOAL PERIODS IN DETAIL

G35 ~ mixed
G36 d descriptive
G37 ✓ supported
G38 ~ mixed
G39 ~ mixed
G41 ~ mixed
G42 × failed
G44 ✓ supported

H107-room-split-onset · generated 2026-10-04 14:59 by in-fra/summaries/build_summaries.py · card hypotheses/H107-room-split-onset/README.md · ratings: H107 agent (suggested) (page 2)